MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 4

(1) Decide if the following statements are true or false. Give a line or two in justification for your answer. It is permissible for that justification to be a reference to previous homeworks or to lectures:

(a) The polynomials $1, x^2, i + ix^2$ in $\mathrm{Poly}_{\leq 2}(\mathbb{C})$ are linearly independent.

Solution. False. We can write $i+ix^2$ as a $\mathbb{C}$-linear combination $i\cdot 1+i\cdot x^2$ of the first two polynomials.

(b) Any set of vectors containing 0 cannot be linearly independent.

Solution. True. This is because we can write a non-trivial linear combination $a \cdot 0 = 0$ where we can choose a to be any non-zero element of the field.

(c) Every subspace of a vector space V is a span of some subset of V.

Solution. True. If $W \subset V$ is a subspace, then it is the smallest subspace containing itself! In other words, $\mathrm{span}(W) = W$.

(d) The intersection of two subspaces of a vector space V is once again a subspace of V.

Solution. True. We showed this on Problem 3 of Homework 3.

(e) If $\{v_1, v_2\}$ is a linearly independent subset of V and $v_3 \notin \mathrm{span}(\{v_1, v_2\})$, then $\{v_1, v_2, v_3\}$ is a linearly independent subset as well.

Solution. True. If we have a linear combination $a_1v_1 + a_2v_2 + a_3v_3 = 0$, then a_3 must be 0, since otherwise, after multiplying by a_3^{-1}, we can write v_3 as a linear combination of v_1 and v_2. But if $a_3 = 0$, then the linear independence of v_1 and v_2 tells us that $a_1 = a_2 = 0$.

(2) Consider the subset

$$U = \{f(x) \in \mathrm{Poly}_{\leq 3}(\mathbb{R}) : \int_0^1 f(x)\, dx = 0\} \subset \mathrm{Poly}_{\leq 3}(\mathbb{R}).$$

(a) Show that U is a subspace of $\mathrm{Poly}_{\leq 3}(\mathbb{R})$.

Solution.
(Contains 0): The integral of the 0 polynomial is clearly 0.
(Closed under $+$): If $f_1(x), f_2(x) \in U$, then we have

$$\int_0^1 f_1(x) + f_2(x)\, dx = \int_0^1 f_1(x)\, dx + \int_0^1 f_2(x)\, dx = 0 + 0 = 0.$$

(Closed under scalar multiplication): If $f(x) \in U$ and $c \in F$, then

$$\int_0^1 (c \cdot f)(x)\, dx = \int_0^1 c \cdot f(x)\, dx = c \cdot \int_0^1 f(x)\, dx = c \cdot 0 = 0.$$

(b) Find a basis for U and extend it to a basis for $\mathrm{Poly}_{\leq 3}(\mathbb{R})$. What is the dimension of U?

Solution. The integral of the polynomial $a_0 + a_1x + a_2x^2 + a_3x^3$ is given by

$$\left[a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} + a_3\frac{x^4}{4}\right]_0^1 = a_0 + \frac{a_1}{2} + \frac{a_2}{3} + \frac{a_4}{4}.$$

Therefore, this polynomial belongs to U precisely when we have

$$a_0 + \frac{a_1}{2} + \frac{a_2}{3} + \frac{a_4}{4} = 0.$$

This tells us that the condition to be in U is
$$a_0 = -\frac{a_1}{2} - \frac{a_2}{3} - \frac{a_4}{4}.$$
the polynomials $-1+2x$, $-1+3x^2$ and $-1+4x^3$ are all in U (why?).
Also, note that
$$\frac{a_1}{2}(-1+2x) + \frac{a_2}{3}(-1+3x^2) + \frac{a_4}{4}(-1+4x^3) = \left(-\frac{a_1}{2} - \frac{a_2}{3} - \frac{a_3}{4}\right) + a_1x + a_2x^2 + a_3x^3.$$
This tells us every element of U is a linear combination of the three given polynomials.
Moreover, these polynomials are linearly independent. One can see this by looking at the result of the linear combination above, but we can also check directly: Suppose that we have a linear combination equaling 0:
$$0 = c_1(-1+2x) + c_2(-1+3x^2) + c_3(-1+4x^3) = (-c_1 - c_2 - c_3) + 2c_1x + 3c_2x^2 + 4c_3x^3.$$
Comparing coefficients now tells us that we must have $0 = c_1 = c_2 = c_3$.
This tells us that $\{-1+2x, -1+3x^2, -1+4x^3\}$ is a basis for U, and in particular U has dimension 3. To extend this to a basis for $\mathrm{Poly}_{\leq 3}(\mathbb{R})$, we just need to add one vector that is not in U (why?), and we can take this to be the constant polynomial 1, for instance.

(3) Consider the subset
$$V = \{f(x) \in \mathrm{Poly}_{\leq 5}(\mathbb{Q}) : \ f(1) = f(2) = 0\} \subset \mathrm{Poly}_{\leq 5}(\mathbb{Q}).$$

(a) Show that V is a subspace of $\mathrm{Poly}_{\leq 5}(\mathbb{Q})$.

Solution.
(Contains 0): The 0 polynomial takes value 0 everywhere, including at 1 and 2.
(Closed under $+$): The sum of two polynomials vanishing at 1 and 2 clearly also vanishes at both values.
(Closed under scalar multiplication): Same for the scalar multiple.

(b) Find a basis for V and extend it to a basis for $\mathrm{Poly}_{\leq 5}(\mathbb{Q})$. What is the dimension of V?

Solution. The given conditions for a polynomial $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ translate to the equations
$$f(1) = a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 0$$
$$f(2) = a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 + 32a_5 = 0$$
This is a system of linear equations and can be solved in the usual way:
$$\left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 16 & 32 & 0 \end{array}\right] \xrightarrow{R2-R1} \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 7 & 15 & 31 & 0 \end{array}\right]$$
$$\xrightarrow{R1-R2} \left[\begin{array}{cccccc|c} 1 & 0 & -2 & -6 & -14 & -30 & 0 \\ 0 & 1 & 3 & 7 & 15 & 31 & 0 \end{array}\right]$$

This now translates to the equations
$$a_0 = 2a_2 + 6a_3 + 14a_4 + 30a_5$$
$$a_1 = -3a_2 - 7a_3 - 15a_4 - 31a_5.$$
So a_2, a_3, a_4, a_5 can be arbitrary and fixing their values determines the values of a_0 and a_1. Setting each of them equal to 1 and the rest 0 in turn, this gives us four polynomials
$$2 - 3x + x^2, 6 - 7x + x^3, 14 - 15x + x^4, 30 - 31x + x^5.$$
These will span V and are linearly independent, and so form a basis. In particular, V has dimension 4. The given basis can be extended to a basis for V by adding back in the polynomials 1 and x.

(4) Consider the subset

$$W = \left\{ A \in M_{2\times 2}(\mathbb{Z}/3\mathbb{Z}) : \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} A = A \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \right\}.$$

(a) Show that W is a subspace of $M_{2\times 2}(\mathbb{Z}/3\mathbb{Z})$.

Solution. Set $B = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$. The condition for being in W is then $AB = BA$. Here, we are taking the matrix product:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{bmatrix}.$$

(Contains 0): If $A = 0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then $0 \cdot B = B \cdot 0 = 0$, so $0 \in W$.
(Closed under $+$): This uses a basic distributive property of matrix multiplication:

$$(A_1 + A_2)B = A_1B + A_2B \; ; \; B(A_1 + A_2) = BA_1 + BA_2.$$

Therefore, if $A_1, A_2 \in W$, then we see that $A_1B + A_2B = BA_1 + BA_2$, and so $A_1 + A_2 \in W$.
(Closed under scalar multiplication): This uses the basic associative property

$$(cA)B = c(AB) \; ; B(cA) = c(BA).$$

Therefore, if $A \in W$, then $c(AB) = c(BA)$, and so $cA \in W$.

(b) Find a basis for W and extend it to a basis for $M_{2\times 2}(\mathbb{Z}/3\mathbb{Z})$. What is the dimension of W?

Solution. Let us write the condition $AB = BA$ explicitly, when $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix} \; ; BA = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a-c & b-d \\ a-c & b-d \end{bmatrix}$$

Therefore, the condition is given by the four equations:

$$\begin{aligned} a+b &= a-c \\ -a-b &= b-d \\ c+d &= a-c \\ -c-d &= b-d \end{aligned}$$

These can be simplified to:

$$\begin{aligned} b+c &= 0 \\ -a+2b+d &= 0 \\ -a-2c+d &= 0 \\ -b-c &= 0. \end{aligned}$$

We can solve this system directly: It comes down to the conditions $b = -c$ and $a = -2c + d$. Since we are in $\mathbb{Z}/3\mathbb{Z}$, we have $-1 = 2$ and $-2 = 1$. So we can write these conditions as $b = 2c$ and $a = c + d$. Therefore, W consists of matrices of the form

$$\begin{bmatrix} c+d & 2c \\ c & d \end{bmatrix}.$$

A basis is given by the matrices with $c = 1, d = 0$ and $c = 0, d = 1$:

$$A_1 = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

To extend this to a basis for $M_{2\times 2}(\mathbb{Z}/3\mathbb{Z})$, we need to add two more matrices so that the four more matrices can together span all of $M_{2\times 2}(\mathbb{Z}/3\mathbb{Z})$ (why is two the magic number here?). Since we were looking at matrices where either c or d are non-zero, we should look to add matrices where they are both zero. We can do this by adding the matrices

$$A_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, A_4 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Note that $A_1 - A_3 - 2A_4 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $A_2 - A_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. This tells us that the 'standard' basis matrices are all in the span of A_1, A_2, A_3, A_4, which means that they must be a basis (why?). The dimension of W is of course 2.

(5) If $f(x)$ is a polynomial of degree n over $\mathbb{R}$, show that the polynomials

$$1, f(x), f'(x), \ldots, f^{(n-1)}(x)$$

form a basis for $\mathrm{Poly}_{\leq n}(\mathbb{R})$.

Solution. Since we already know that $\mathrm{Poly}_{\leq n}(\mathbb{R})$ has dimension $n+1$, it suffices to know that these polynomials are linearly independent; see Observation 3 from Lecture 12.

For this, the following generalization of 1(e) will be helpful: If $v_1, \ldots, v_n$ is a list of vectors such that $v_1 \neq 0$ and such that, for each $j = 2, \ldots, n$, v_j does *not* belong to the span of $v_1, \ldots, v_{j-1}$, then $v_1, \ldots, v_n$ are linearly independent.

We can prove this by induction on n: The base case of $n = 1$ is essentially saying that v_1 is non-zero, which is our assumption.

The inductive hypothesis is that the assertion is true for $n-1$ vectors. In particular, the list $v_1, \ldots, v_{n-1}$ is linearly independent. If we had an equation

$$a_1 v_1 + \cdots + a_{n-1} v_{n-1} + a_n v_n = 0,$$

and if $a_n \neq 0$, then we could multiply by a_n^{-1} and move the terms involving $v_1, \ldots, v_{n-1}$ to the other side of the equation. This would exhibit v_n as a linear combination of the first $n-1$ vectors, which is a contradiction. Therefore, $a_n = 0$. But then we have a linear combination of the first $n-1$ vectors equating to 0. By our inductive hypothesis, this is only possible if $a_1 = a_2 = \cdots = a_{n-1} = 0$.

In our particular case, it is easier to apply the criterion to the rearranged list of $n+1$ vectors

$$v_1 = 1, v_2 = f^{(n-1)}(x), v_3 = f^{(n-2)}(x), \ldots, v_{n-1} = f'(x), v_{n+1} = f(x).$$

It is clear that at each stage v_j is not in the span of $v_1, \ldots, v_{j-1}$. Indeed, this span can only involve polynomials of degree $\leq j-2$, while v_j has degree $j-1$. This shows that the list is linearly independent and finishes the proof.

(6) Compute the change of basis matrices (in both directions) between the bases

$$\mathcal{B}_1 = \{1, x, x^2, x^3\}\ ;\ \mathcal{B}_2 = \{1, x-2, (x-2)^2, (x-2)^2, (x-2)^3\}$$

for the $\mathbb{Z}/5\mathbb{Z}$-vector space $\mathrm{Poly}_{\leq 3}(\mathbb{Z}/5\mathbb{Z})$.

Solution. Recall that the change of basis matrix from a basis $\mathcal{B}_1 = \{v_1, \ldots, v_n\}$ to another basis $\mathcal{B}_2 = \{w_1, \ldots, w_n\}$ is the matrix whose entries in the i-th row and the j-th column are determined by:

$$w_j = a_{1,j} v_1 + \cdots + a_{n,j} v_n.$$

In other words, the j-th column of the matrix consists of the coefficients of w_j written in terms of the basis $\mathcal{B}_1$.

In the lecture, I messed up the rows and columns, so if you ended up with matrices with the rows and columns flipped, that's okay for this homework!

We have

$$\begin{aligned}
1 &= 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 + 0 \cdot x^3 \\
x - 2 = x + 3 &= 3 \cdot 1 + 1 \cdot x + 0 \cdot x^2 + 0 \cdot x^3 \\
(x-2)^2 = x^2 - 2x + 4 = x^2 + 3x + 4 &= 4 \cdot 1 + 3 \cdot x + 1 \cdot x^2 + 0 \cdot x^3 \\
(x-2)^3 = x^3 - 6x^2 + 12x - 8 = x^3 + 4x^2 + 2x + 2 &= 2 \cdot 1 + 2 \cdot x + 4 \cdot x^2 + 1 \cdot x^3
\end{aligned}$$

Therefore, the change of basis matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$ is

$$\begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

To compute the matrix in the other direction, we note:

$$\begin{aligned}
1 &= 1 \cdot 1 + 0 \cdot (x-2) + 0 \cdot (x-2)^2 + 0 \cdot (x-2)^3 \\
x &= 2 \cdot 1 + 1 \cdot (x-2) + 0 \cdot (x-2)^2 + 0 \cdot (x-2)^3 \\
x^2 = -4 + 4x + (x-2)^2 = (-4-2) - (x-2) + (x-2)^2 &= 4 \cdot 1 + 4 \cdot (x-2) + 1 \cdot (x-2)^2 + 0 \cdot (x-2)^2 \\
x^3 = 3 + 3(2 + (x-2)) + (4 + 4(x-2) + (x-2)^2) + (x-2)^3 &= 3 \cdot 1 + 2 \cdot (x-2) + 4 \cdot (x-2)^2 + 1 \cdot (x-2)^3
\end{aligned}$$

Therefore, the change of basis matrix in the other direction is

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$